

Student Record System

Test Plan — 50 Test Cases

FR-001: Record Attendance

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_001_01	Attendance	Test valid attendance recording	Faculty logged in; student and subject exist	Open Attendance → select subject/student → mark Present → Submit	PES1UG24CS647; DBMS; Present	Attendance is saved and confirmation displayed.	Not executed — to be filled after manual testing.	Pending
UT_001_02	Attendance	Test invalid student ID	Faculty logged in	Open Attendance → enter ID → submit	INVALID123; Present	Invalid ID is rejected with an error.	Not executed — to be filled after manual testing.	Pending
UT_001_03	Attendance	Test missing attendance status	Faculty logged in; valid student selected	Select student/subject → leave status blank → Submit	PES1UG24CS625; blank	Submission is prevented and status-required message appears.	Not executed — to be filled after manual testing.	Pending
IT_001_04	Attendance & Database	Test storage and retrieval of attendance	Valid student record exists	Mark Present → Save → reopen attendance	PES1UG24CS627; Present	Saved attendance is retrieved correctly.	Not executed — to be filled after manual testing.	Pending
IT_001_05	Attendance & Student Profile	Test attendance appears in student profile	Faculty and student accounts exist	Faculty records attendance → student logs in → opens Attendance	PES1UG24CS647; Present	New attendance appears in student profile.	Not executed — to be filled after manual testing.	Pending
IT_001_06	Attendance	Test updating attendance	Existing record; faculty authorized	Open record → change Present to Absent → Save → reopen	Old: Present; New: Absent	Updated status is saved and displayed.	Not executed — to be filled after manual testing.	Pending
ST_001_07	Attendance	Test attendance for multiple students	Multiple students registered	Select subject → mark three students → submit → review	647 Present; 625 Absent; 627 Present	Each record is saved against the correct student.	Not executed — to be filled after manual testing.	Pending
ST_001_08	Attendance	Test duplicate attendance prevention	Record already exists for same class	Open same class → submit another record	PES1UG24CS647; same subject/date/class	Duplicate entry is prevented with a message.	Not executed — to be filled after manual testing.	Pending
ST_001_09	Attendance	Test unauthorized attendance entry	Student account without faculty privilege	Student logs in → attempts attendance entry	Role: Student	Access to attendance entry is denied.	Not executed — to be filled after manual testing.	Pending
ST_001_10	Attendance	Test persistence after relogin	Attendance already saved	Record attendance → logout → login → view attendance	PES1UG24CS625; Present	Saved attendance remains available.	Not executed — to be filled after manual testing.	Pending

FR-002: Calculate Attendance Percentage

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_002_01	Attendance Calculation	Test percentage with valid values	Valid attendance counts exist	Enter attended/total → Calculate	8/10	80% is displayed.	Not executed — to be filled after manual testing.	Pending
UT_002_02	Attendance Calculation	Test 100% attendance	Valid counts	Enter attended/total → Calculate	10/10	100% is displayed.	Not executed — to be filled after manual testing.	Pending
UT_002_03	Attendance Calculation	Test 0% attendance	Valid total exists	Enter attended/total → Calculate	0/10	0% is displayed.	Not executed — to be filled after manual testing.	Pending
UT_002_04	Attendance Calculation	Test attended greater than total	Calculation screen available	Enter values → Calculate	12/10	Values are rejected with validation message.	Not executed — to be filled after manual testing.	Pending
IT_002_05	Attendance & Database	Test update after new attendance	Existing 7/10 record	Add Present class → save → view percentage	7/10 → 8/10	Percentage updates to 80%.	Not executed — to be filled after manual testing.	Pending
IT_002_06	Attendance & Profile	Test percentage in profile	Attendance records exist	Open profile → Attendance	9/12	75% is displayed.	Not executed — to be filled after manual testing.	Pending
ST_002_07	Attendance Calculation	Test multiple subjects	Multiple attendance records exist	Open summary → review each subject	DBMS 8/10; OS 9/12; CN 7/10	80%, 75%, 70% displayed respectively.	Not executed — to be filled after manual testing.	Pending
ST_002_08	Attendance Calculation	Test zero total classes	Calculation screen available	Enter 0/0 → Calculate	0/0	No divide-by-zero error; defined message/value displayed.	Not executed — to be filled after manual testing.	Pending
ST_002_09	Attendance Calculation	Test negative values	Calculation screen available	Enter negative attended value → Calculate	-2/10	Negative value is rejected.	Not executed — to be filled after manual testing.	Pending
ST_002_10	Attendance Calculation	Test non-numeric input	Calculation screen available	Enter text → Calculate	eight/10	Non-numeric input is rejected.	Not executed — to be filled after manual testing.	Pending

FR-003: Enter and Update Marks

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_003_01	Marks	Test valid mark entry	Faculty logged in; student/subject exist	Open Marks → select student/subject → enter → Save	PES1UG24CS647; 85/100	Marks saved and confirmation displayed.	Not executed — to be filled after manual testing.	Pending
UT_003_02	Marks	Test marks above maximum	Faculty logged in	Enter mark → Save	110/100	Mark is rejected.	Not executed — to be filled after manual testing.	Pending
UT_003_03	Marks	Test negative marks	Faculty logged in	Enter negative mark → Save	-5/100	Negative mark is rejected.	Not executed — to be filled after manual testing.	Pending
UT_003_04	Marks	Test blank marks	Faculty logged in; student selected	Leave mark blank → Save	Blank	Submission is prevented and required-field message appears.	Not executed — to be filled after manual testing.	Pending
IT_003_05	Marks & Database	Test storage and retrieval	Valid student/subject exist	Enter 78 → Save → reopen record	PES1UG24CS625; 78/100	78/100 is retrieved correctly.	Not executed — to be filled after manual testing.	Pending
IT_003_06	Marks & Grade	Test grade update after mark change	Grading scheme configured	Change mark → Save → open grade	65 → 85	Grade updates according to grading scheme.	Not executed — to be filled after manual testing.	Pending
ST_003_07	Marks	Test updating existing mark	Existing mark; faculty authorized	Open mark → change → Save → reopen	PES1UG24CS627; 70 → 82	82 is displayed.	Not executed — to be filled after manual testing.	Pending
ST_003_08	Marks	Test unauthorized mark modification	Student account exists	Student logs in → attempts edit/save	Role: Student; 95	Modification is denied.	Not executed — to be filled after manual testing.	Pending
ST_003_09	Marks	Test separate subjects	Student has multiple subjects	Enter marks for two subjects → view	DBMS 85; OS 78	Each mark appears under the correct subject.	Not executed — to be filled after manual testing.	Pending
ST_003_10	Marks	Test non-numeric mark	Faculty logged in	Enter text → Save	eighty	Input is rejected.	Not executed — to be filled after manual testing.	Pending

FR-004: Calculate GPA

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_004_01	GPA	Test GPA with valid grades and credits	Valid grades/credits exist	Open GPA → select semester → Calculate	Grade points 8,9,7; credits 3,4,3	Correct weighted GPA is displayed.	Not executed — to be filled after manual testing.	Pending
UT_004_02	GPA	Test equal grade points	Valid grades/credits	Enter/select grades → Calculate	8,8,8; credits 3,4,3	GPA is 8.00.	Not executed — to be filled after manual testing.	Pending
UT_004_03	GPA	Test missing grade	Incomplete semester record	Leave one grade blank → Calculate	8, blank, 7	Calculation is prevented or missing-data message shown.	Not executed — to be filled after manual testing.	Pending
UT_004_04	GPA	Test invalid credits	GPA screen available	Enter negative credit → Calculate	Grade point 8; credit -3	Invalid credit is rejected.	Not executed — to be filled after manual testing.	Pending
IT_004_05	Marks & GPA	Test GPA uses latest grade	Marks saved and grading scheme exists	Update mark → Save → open GPA → recalculate	PES1UG24CS647; updated mark 90	GPA reflects latest grade.	Not executed — to be filled after manual testing.	Pending
IT_004_06	GPA & Academic Record	Test GPA storage	Valid semester grades/credits	Calculate GPA → save → reopen record	PES1UG24CS625; semester 3	GPA is stored in semester record.	Not executed — to be filled after manual testing.	Pending
ST_004_07	GPA	Test multiple subjects/credits	All grades/credits valid	Select semester → calculate	9,8,7,10; credits 3,4,3,2	Weighted GPA is calculated correctly.	Not executed — to be filled after manual testing.	Pending
ST_004_08	GPA	Test unauthorized GPA modification	Student account exists	Student logs in → attempts edit GPA	Role: Student; 10.00	Unauthorized modification is prevented.	Not executed — to be filled after manual testing.	Pending
ST_004_09	GPA	Test recalculation after grade update	Existing GPA and editable grade	Change grade → save → recalculate	7 → 9 grade point	GPA updates correctly.	Not executed — to be filled after manual testing.	Pending
ST_004_10	GPA	Test GPA display precision	Valid GPA calculated	Open GPA record → review format	Calculated value 8.375	GPA follows defined precision, e.g. 8.38 if two decimals.	Not executed — to be filled after manual testing.	Pending

FR-005: View Academic Records

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_005_01	Student Profile	Test viewing personal information	Valid student account/profile	Login → open Profile	PES1UG24CS647	Personal and academic information is displayed.	Not executed — to be filled after manual testing.	Pending
UT_005_02	Student Login	Test invalid password	Registered student exists	Enter valid ID + wrong password → Login	PES1UG24CS625; wrong password	Login is denied with an error.	Not executed — to be filled after manual testing.	Pending
UT_005_03	Academic Records	Test viewing marks and grades	Student logged in; records exist	Open Academic Records → select semester	PES1UG24CS627; semester 3	Subject-wise marks and grades are displayed.	Not executed — to be filled after manual testing.	Pending
UT_005_04	Academic Records	Test semester with no records	Student logged in; selected semester has no data	Open Academic Records → select semester	PES1UG24CS647; semester 1	No-records message is displayed.	Not executed — to be filled after manual testing.	Pending
IT_005_05	Academic Records & Attendance	Test attendance visibility	Attendance data saved	Login → open Attendance/Records	PES1UG24CS625; DBMS	Saved attendance is displayed.	Not executed — to be filled after manual testing.	Pending
IT_005_06	Academic Records & GPA	Test GPA/CGPA visibility	GPA/CGPA calculated	Login → open Academic Records	PES1UG24CS627	Correct GPA and CGPA are displayed.	Not executed — to be filled after manual testing.	Pending
ST_005_07	Academic History	Test multiple semester records	Records exist for several semesters	Login → open history → select semesters 1,2,3	PES1UG24CS647; semesters 1–3	Correct records display for each semester.	Not executed — to be filled after manual testing.	Pending
ST_005_08	Academic Records	Test access to another student's record	Two student accounts exist	Student A logs in → attempts Student B record	A: PES1UG24CS647; B: PES1UG24CS625	Access is denied.	Not executed — to be filled after manual testing.	Pending
ST_005_09	Academic Records	Test records after relogin	Academic records exist	Login → view → logout → login → view	PES1UG24CS625	Authorized records remain available.	Not executed — to be filled after	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
							manual testing.	
ST_005_10	Academic Records	Test database unavailable handling	Database unavailability can be simulated	Simulate DB unavailable → open records	Database: Unavailable	Clear error is shown; corrupted/misleading data is not displayed.	Not executed — to be filled after manual testing.	Pending